Lorenzo Maggi

Biomedical Engineer

Hands-on biomedical engineer (PhD) with experience in **surgical robotics, knee biomechanics, and ex-vivo experimentation.** During my PhD I developed a knee kinematic simulator and contributed to the design of **Robin®**, a robotic system for total knee arthroplasty.

Strong **maker** mindset **(mechanics, electronics, 3D printing)** and a focus on **translational** research.

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https://lorenzo-maggi.github.io/

Experience & Education

R&D Robotics engineer

November 2025 - ongoing

Orthokey Italia Srl

- **Co-development** of **Robin®** robotic system for **TKA**;
- **ISO 13485, IEC 60601, IEC 80601** oriented design;
- **KUKA LBR Med** software development;
- Collaboration with **National Research Council** for Murine emotional analysis via **neural networks**.

PhD in Industrial Engineering

November 2022 - October 2025

Orthokey Italia Srl - University of Florence

- **Development** of an innovative **knee laxity** assessment robotic tool based on **KUKA LBR Med**;
- **Development** of a **force-feedback control** for rehabilitation **hand exoskeleton**;
- **2 years** of Industrial experience at **Orthokey Italia Srl**;
- A brief **summary** of my **PhD**: https://youtu.be/5DqW-fMwWBU

Visiting PhD researcher

September 2024 - March 2025

Université Libre de Bruxelles (Prof. B. Innocenti)

- **6-month abroad experience**;
- **Development** of a portable **knee kinematic** simulator for kinematic analysis of different **TKA implants**;
- Implementation of a **PID controller** to simulate **quadriceps tendon loading**;
- Participation in several **cadaveric knee tests** with the kinematic simulator.

MSc in Biomedical engineering (110/110 cum laude)

October 2022

BSc in Mechanical engineering

February 2021

University of Florence

Languages

Italian (mother tongue)

English (fluent)

Spanish (beginner)

Core skills

Robotics & control: **KUKA LBR Med**;
Software: **Python, C/C++, Java, MATLAB, Arduino, SolidWorks**.

Lab skills: **sensor** calibration **soldering** (incl. SMD), **3D printing** (FDM), general **mechanical skills**.

Licence of professional engineer.
Scientific writing.
Knee biomechanics.
Driving licence: A, B.

Hobbies & Other activities

Volunteering**:**

- **Ambulance driver** and first **responder** (2014 - present);
- **BLSD and safety trainer** for internal and external staff;
- Event **organizer** and **cook** (with HACCP) for **country fair** (**Fa'Festa** 2011 - present).

Personal projects:

- **DIY sous-vide cooker**;
- **Programmable PSU**
- **Full restoration** of a **Piaggio Si**.

Interests:

- **Trekking and mountain biking;**
- **Cooking**.

I hereby authorize the processing of my personal data in accordance with Legislative Decree No. 196 of June 30, 2003 and Article 13 of the EU General Data Protection Regulation (GDPR).